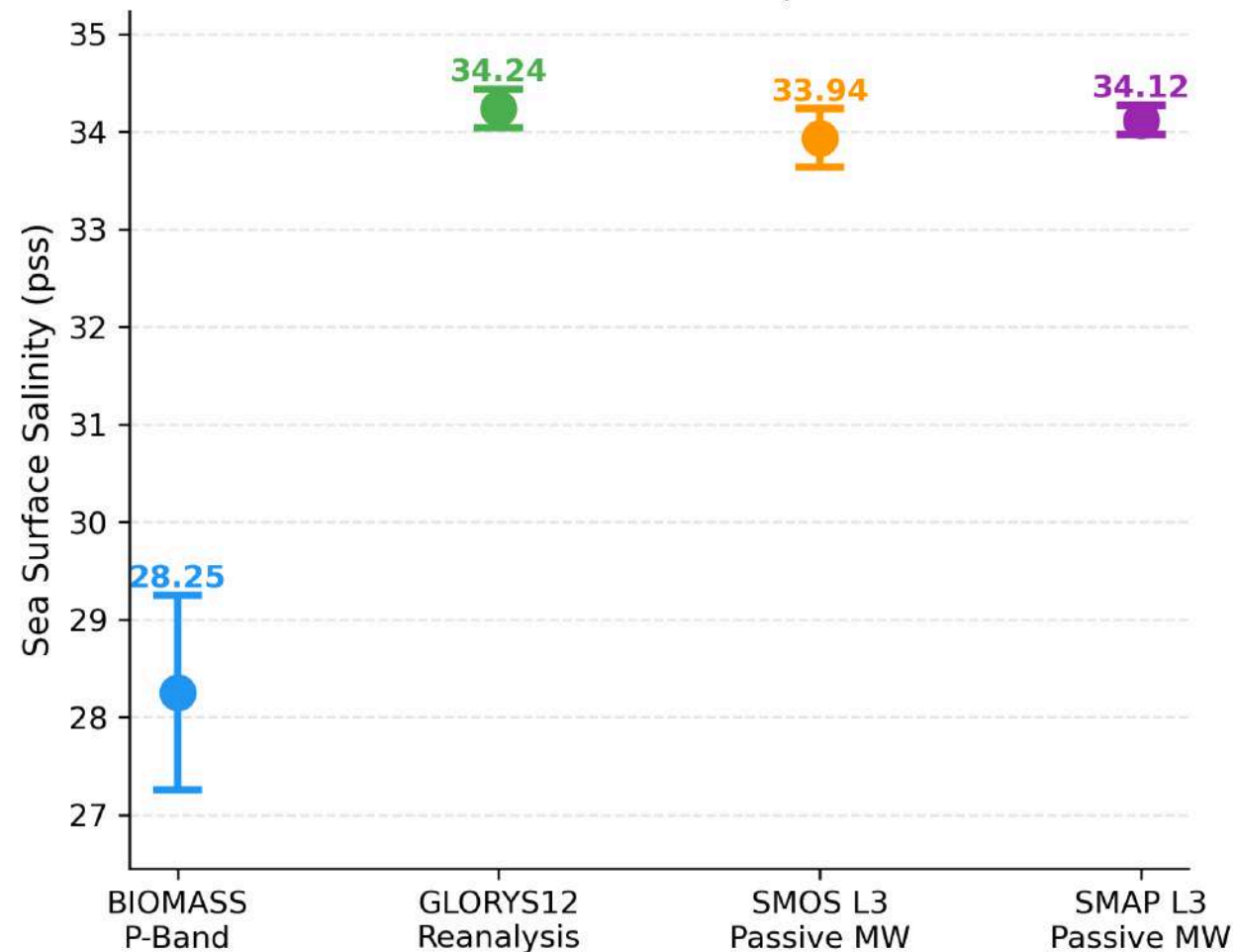


BIOMASS P-Band Sea Surface Salinity Validation

Scene Mean SSS Comparison



Summary

Location : 13.90°S, 89.64°E
Date : 2026-05-08
Incidence : 25.6°
U10 (ERA5) : 7.8 m/s

SSS Values (pss)

BIOMASS : 28.25 ± 1.0
GLORYS12 : 34.24 ± 0.2
SMOS L3 : 33.94 ± 0.3
SMAP L3 : 34.12 ± 0.1

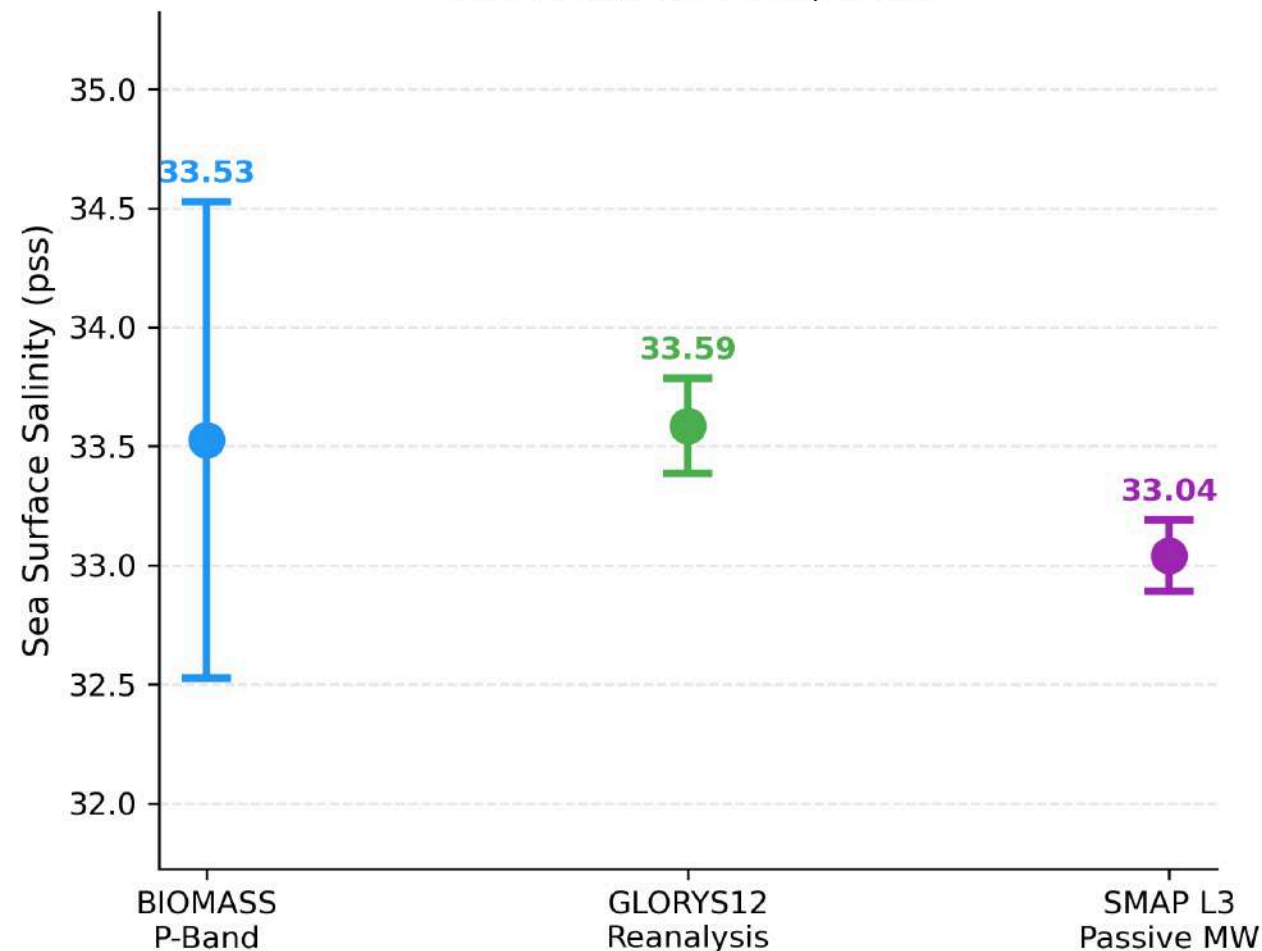
Differences (pss)

BIO - GLORYS : -5.99
BIO - SMOS : -5.69
BIO - SMAP : -5.87
SMOS - GLORYS : -0.30
SMAP - GLORYS : -0.12
SMOS - SMAP : -0.18

Valid pixels : 31,918,317
SMOS dist : 0.1° (~12 km)
SMAP dist : 0.1° (~7 km)

BIOMASS P-Band Sea Surface Salinity Validation

Scene Mean SSS Comparison



Summary

Location : 36.79°S, 53.53°W
Date : 2026-05-05
Incidence : 25.3°
U10 (ERA5) : 5.7 m/s

SSS Values (pss)

BIOMASS : 33.53 ± 1.0
GLORYS12 : 33.59 ± 0.2
SMOS L3 : nan ± 0.3
SMAP L3 : 33.04 ± 0.1

Differences (pss)

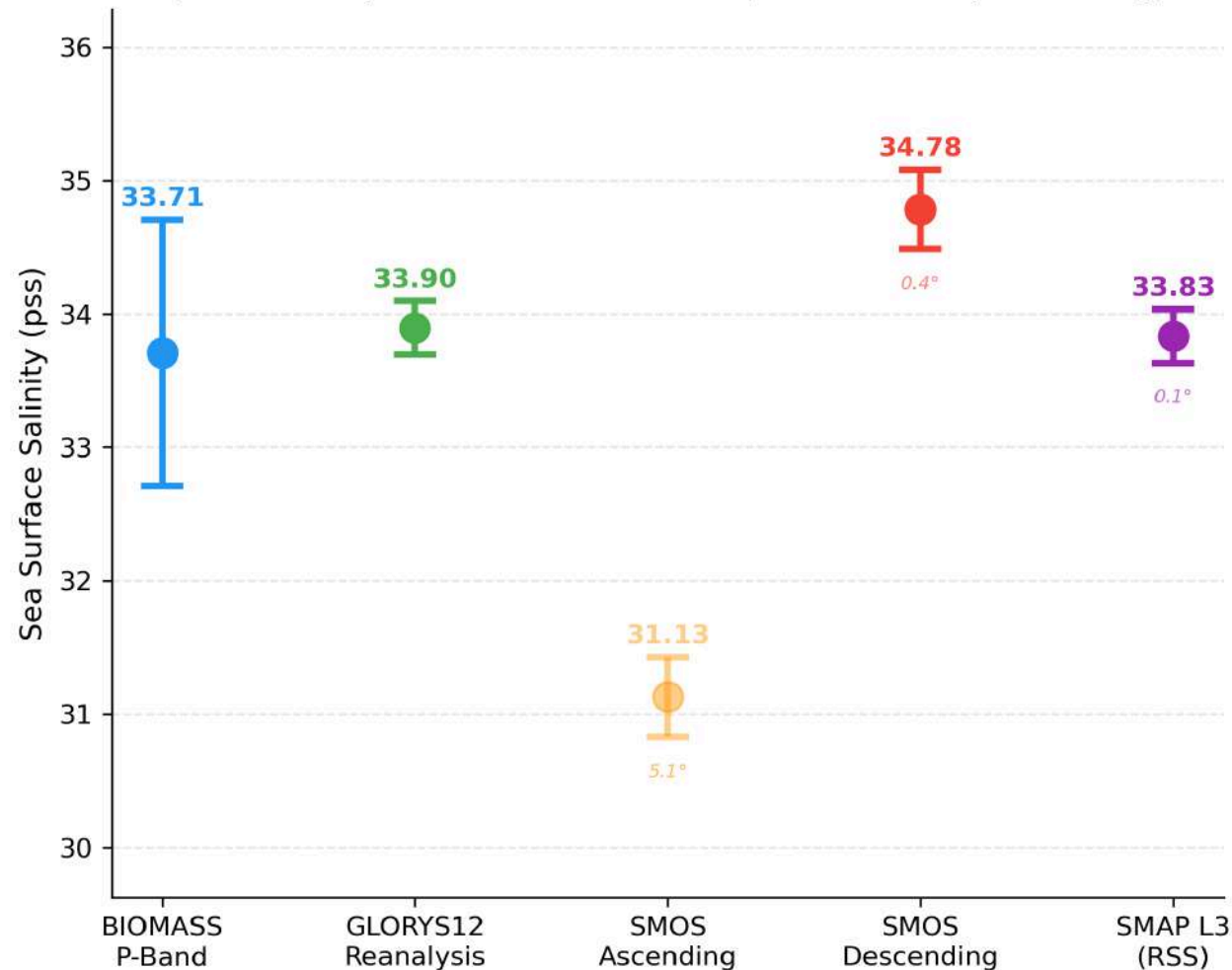
BIO - GLORYS : -0.06
BIO - SMOS : n/a
BIO - SMAP : +0.49
SMOS - GLORYS : n/a
SMAP - GLORYS : -0.55
SMOS - SMAP : n/a

Valid pixels : 22,329,838
SMOS dist : nan° (~nan km)
SMAP dist : 0.0° (~4 km)

BIOMASS P-Band Sea Surface Salinity Validation

Scene Mean SSS Comparison

(BIOMASS = empirical, anchored to GLORYS12 | faded = nearest pixel >1° away)



Summary

Location : 51.83°S, 57.36°W
Date : 2026-03-01
Incidence : 25.3°
U10 : 5.3 m/s
Ocean : 100.0%
Coherence : 0.9854

Sea Surface Temperature

GLORYS12 centre : 10.16 °C [T prior]
GLORYS12 bbox : 9.63 ± 2.30 °C
SST→SSS impact : ±2°C → ~±0.15 pss

SSS Values (pss)

BIOMASS : 33.71 ± 1.00
[empirical, GLORYS anchor]
GLORYS12 : 33.90 ± 0.20
SMOS asc : 31.13 ± 0.30 Δ 5.1°
SMOS des : 34.78 ± 0.30 (0.40°)
SMAP L3 : 33.83 ± 0.20 (0.05°)

Differences vs GLORYS12 (pss)

BIO : -0.19
SMOS asc : -2.77
SMOS des : +0.89
SMAP : -0.06

Passive MW inter-sensor (pss)

SMOS asc - SMAP : -2.70

Valid pixels : 607,076 (2.9%)

Δ = nearest pixel >1° (swath gap)